

OMAR NOURI

Senior Data Scientist | AI / Machine Learning Engineer | Generative AI, LLM & RAG Systems
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PROFESSIONAL SUMMARY

Senior Data Scientist and AI/ML Engineer with 5+ years of experience designing and deploying production-grade Generative AI, NLP, and machine learning systems for enterprise clients in banking, insurance, and fintech. Deep hands-on expertise in Retrieval-Augmented Generation (RAG), LLM integration, agentic AI workflows, prompt engineering, and cloud-based MLOps on AWS and GCP. Proven track record improving system accuracy, reducing latency, and scaling data pipelines processing hundreds of thousands of records, with production deployments serving 10,000+ users. Open to remote, hybrid, and international (EOR) roles.

CORE SKILLS

Generative AI and LLMs: GPT-4, LLaMA, Claude, Mistral, LangChain, RAG (Retrieval-Augmented Generation), Prompt Engineering, Agentic AI Workflows, Vector Search, Semantic Embeddings
Machine Learning and Deep Learning: Machine Learning, Deep Learning, PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers, MLOps, MLflow, AWS SageMaker
Natural Language Processing: Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Summarization, Translation, Seq2Seq, Language Modeling, Conversational AI (RASA), spaCy, NLTK
Document Intelligence and Computer Vision: Amazon Textract, OCR, Intelligent Document Processing (IDP), OpenCV, Image and Graph Extraction, CNN (Convolutional Neural Networks)
Cloud and Infrastructure: AWS (Lambda, S3, SageMaker, Textract), Google Cloud Platform (GCP), Docker, Apache Airflow, Git, GitHub, Bitbucket, CI/CD
Data Engineering: ETL Pipelines, Data Pipelines, Web Scraping, Data Cleaning and Transformation, Data Governance, Data Privacy and Security, SQL
Databases: MySQL, PostgreSQL, MongoDB, SQLite, Neo4j
Programming Languages: Python, SQL, JavaScript
BI and Visualization: Power BI, Tableau, QlikView
Leadership and Delivery: Technical Advisory, Client Consulting, Cross-Functional Collaboration, Remote Collaboration, Agile / Scrum, Team Mentoring

PROFESSIONAL EXPERIENCE

Senior Data Scientist / AI Consultant, DXC Technology

April 2025 - Present

Rabat, Morocco (Hybrid)

- Lead AI/ML advisory engagements for enterprise insurance clients, defining Generative AI roadmaps and agentic AI architecture strategy.
- Architected and optimized a production Retrieval-Augmented Generation (RAG) system with hybrid search (dense and sparse retrieval) and cross-encoder reranking, reducing response latency by 10 seconds and improving output accuracy by 20 percent.
- Designed and deployed agentic AI workflows on AWS (Lambda, S3), delivering proofs of concept that translated client business requirements into scalable automation solutions.
- Built an Intelligent Document Processing (IDP) pipeline using Amazon Textract for PII extraction across two production systems, combined with LLM-based image and graph data extraction.
- Designed and implemented an Excel parsing and semantic embedding pipeline to feed structured data into downstream RAG workflows.
- Advised stakeholders on adoption of frontier LLM techniques (GPT-4, open-source models), shaping enterprise Generative AI strategy.

Data Scientist / AI Consultant, DXC Technology

January 2024 - April 2025

Rabat, Morocco (Hybrid)

- Consulted on generative AI strategy for enterprise insurance clients, translating business requirements into LLM-based solutions that improved decision-making workflows and customer interaction quality.
- Developed and maintained production MLOps pipelines for model deployment, monitoring, and versioning on AWS.
- Implemented OpenCV-based preprocessing for document and image pipelines feeding downstream extraction models.
- Conducted applied research on frontier LLM techniques (GPT-4, open-source models), advising stakeholders on integration strategy.

Data Scientist, AI and NLP Projects, Digital Place

January 2022 - December 2023

Tangier, Morocco (On-site)

- Won a national fintech challenge by designing and delivering a Darija-dialect conversational banking assistant, fine-tuning Meta Wav2Vec for Darija speech recognition and building an action-based assistant using RASA, demoed to clients at GITEX.
- Built a CNN-based classifier on MFCC audio features for Darija voice command recognition within the voice bot pipeline.
- Deployed French and Arabic NLP transcription models achieving 95 percent accuracy, integrated into a banking application serving 10,000+ active users.

- Built Airflow-based ETL pipelines processing 500,000+ records, applying BERT, GPT, and LLaMA for classification, skill extraction, and semantic recommendations.
- Developed and maintained backend systems supporting data science and ML applications, collaborating with data science teams to productionize models into scalable services.
- Designed MySQL and Neo4j data architectures and built automated Power BI dashboards for executive reporting.

Data Engineer Intern, Digital Place

March 2021 - September 2021

Tangier, Morocco

- Built the first version of automated web scraping and data pipelines covering 100,000+ Moroccan job listings; designed the underlying MySQL schema for structured storage.
- Collaborated with cross-functional teams to translate business requirements into data-driven solutions.

KEY PROJECTS

RAG Insurance AI Assistant (DXC Technology, 2024-Present): End-to-end Generative AI solution on AWS for insurance document Q&A using hybrid retrieval and cross-encoder reranking.

Job Market Intelligence Platform (Prodige/Mouakaba, 2022): Automated pipelines classifying 500,000+ job postings with transformer models; Power BI dashboards for labor market insights.

AI-Enhanced Startup Search Engine (2022): Fine-tuned an attention-based RNN to generate semantic keywords from startup descriptions, improving search relevance.

EDUCATION

State Engineer Diploma (MSc), Computer Engineering and Information Systems. University of Abdelmalek Essaadi, Ecole Nationale des Sciences Appliquees de Tanger (ENSA), Tangier, Morocco. 2015 - 2021.

Baccalaureate Diploma, Mathematical Sciences (Option B). AL Amana Institute, Tangier, Morocco. 2015.

CERTIFICATIONS

Scrum Foundation Professional Certificate (SFPC)

DALF C1, French Language Proficiency

LANGUAGES

Arabic: Native. French: Professional Proficiency (DALF C1). English: Professional Proficiency. Spanish: Elementary.